

NLP Assessment

Customer Feedback Cleaning, Sentence Chunking & Sentiment Analysis

Domain: Consumer Electronics – Smart TV Product Analysis

1. Assessment Objective

You are working as a **Data Analyst / NLP Engineer** for a consumer electronics company.

The company has collected customer feedback for **six different Smart TV products**.

However, the customer feedback is available in raw and unclean formats. The data contains:

- HTML tags
- HTML formatting
- Email addresses
- URLs
- Emojis
- Special characters
- Extra spaces
- Multiple newline characters
- HTML entities
- Repeated punctuation
- Unwanted symbols
- Mixed formatting

Your task is to build a complete **NLP text processing and sentiment analysis pipeline**.

You must:

1. Read the raw customer feedback.
2. Clean the unstructured text.
3. Remove unwanted HTML tags, emails, URLs, emojis, and special characters.
4. Normalize whitespace and punctuation.
5. Perform sentence-level chunking.
6. Assign a unique ID to every sentence chunk.
7. Perform sentiment analysis on every chunk.
8. Calculate the sentiment score for every chunk.

9. Assign a sentiment label:
 10. Positive
 11. Negative
 12. Neutral
 13. Store the sentence-level results in a CSV file.
 14. Generate product-wise sentiment statistics.
 15. Compare all six products.
 16. Identify the best Smart TV based on customer feedback.
 17. Provide a business justification for your recommendation.
-

2. Business Problem

The company wants to understand customer opinions about six competing Smart TV products.

The six products are:

1. TechNova SmartView
2. VisionMax
3. UltraView Pro
4. EcoScreen
5. CineMax Ultra
6. SmartVision X

Your final objective is to answer:

Which Smart TV is the best product based on customer feedback and sentiment analysis?

Your recommendation should be based on the analysis of the customer feedback.

You should consider:

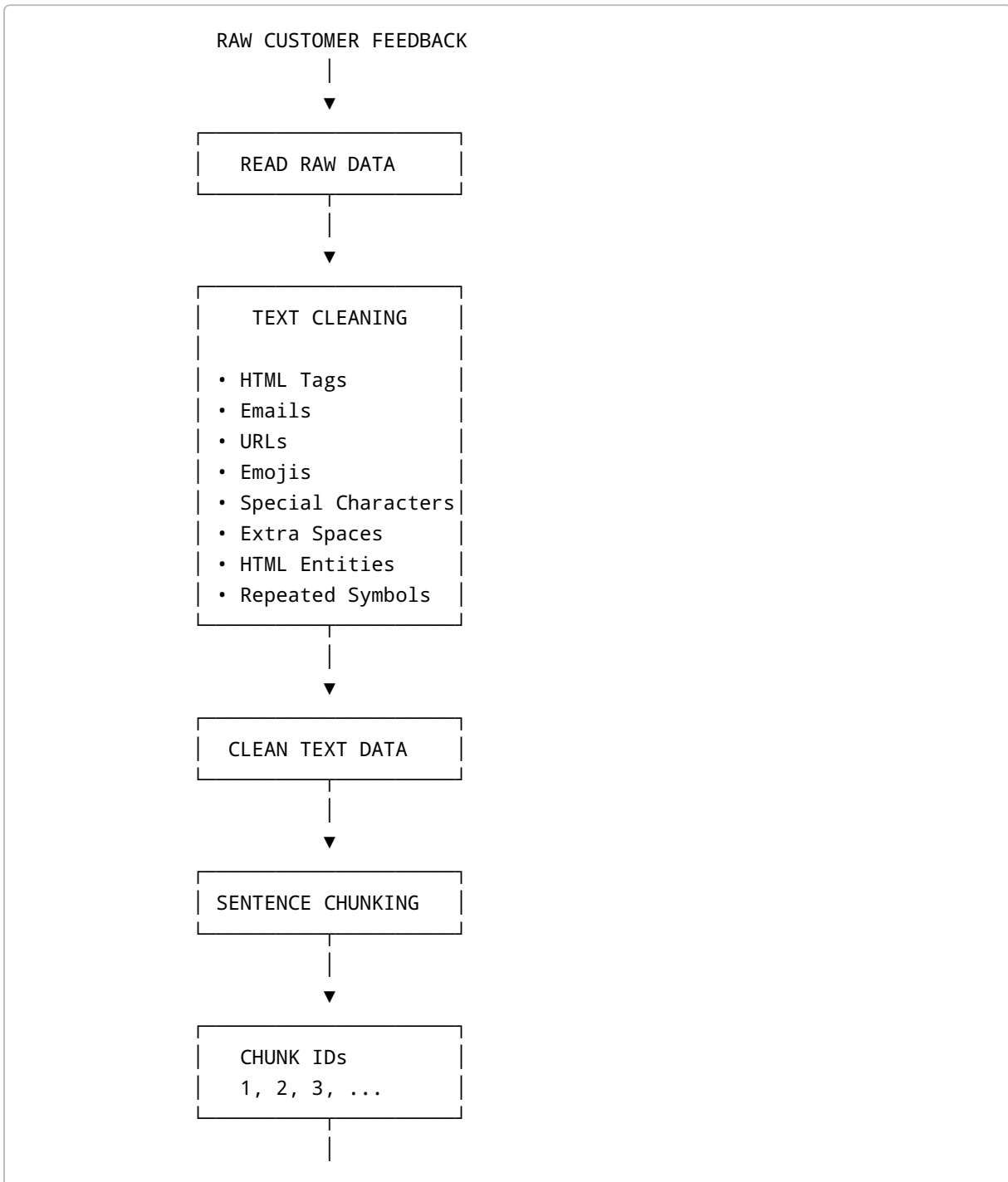
- Overall sentiment
- Average sentiment score
- Number of positive comments
- Number of negative comments
- Number of neutral comments
- Product strengths
- Product weaknesses
- Recurring customer complaints

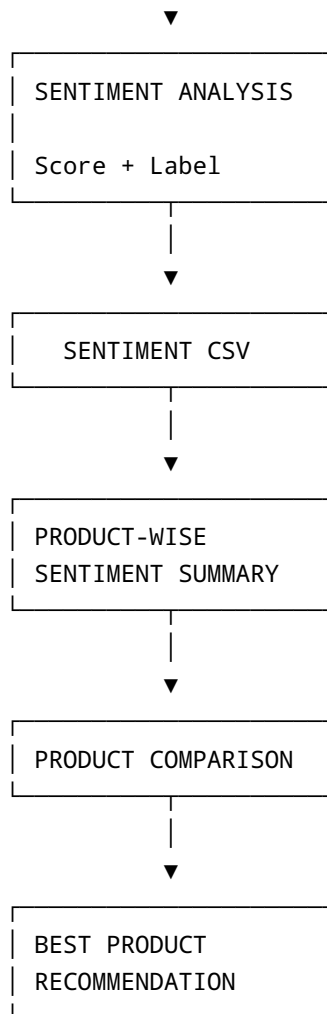
You should not simply select the product with the highest number of positive sentences.

Your recommendation should include a meaningful business justification.

3. Complete NLP Pipeline

Your solution should follow this overall workflow:





4. Input Dataset

The following six customer feedback documents represent the raw input data.

The data is intentionally noisy and contains different types of unclean content.

Students should NOT manually clean the data.

The cleaning process must be implemented programmatically.

PRODUCT 1: TechNova SmartView 55-inch TV

<h1>Customer Feedback - TechNova Electronics</h1>

<p>TechNova Electronics has recently launched its new SmartView 55-inch television, and customers have shared mixed opinions about the product. 😊</p>

<p>The television has a modern design and looks very attractive in the living room!!! The picture quality is excellent, and the colors are bright & realistic. Many customers were also impressed with the sound quality of the television. 👍👍</p>

<p>However, some customers reported problems with the initial setup process. The installation instructions were <i>not very clear</i>, and a few customers found it difficult to connect the television to their Wi-Fi network. 😞</p>

<p>For technical support, customers can contact support@technova-example.com or visit <https://www.technova-example.com/support>.</p>

<p>The remote control is comfortable to use, but the buttons sometimes do not respond immediately. This can be frustrating when users are trying to change channels quickly!!! 😡</p>

<p>The SmartView television supports several popular streaming applications. Customers appreciated being able to watch movies and shows without connecting additional devices. The streaming performance was generally smooth, and the applications loaded quickly. 👍</p>

<p>Unfortunately, some users experienced occasional buffering when watching high-resolution videos... 😞</p>

The display quality is another positive aspect of the product. Customers who purchased the television for watching sports were particularly satisfied with the sharp images. ⚽🏏 The television also performs well when playing video games because the response time is low.

However, a few gamers complained that the television does not have enough HDMI ports for connecting multiple gaming devices. 😡😡

<p>The customer service experience received both positive and negative feedback.</p>

Some customers said that the support team was friendly and helpful when they contacted them. The support representatives explained the solutions clearly and resolved technical issues quickly. 😊😊

On the other hand, some customers had to wait several days before receiving a response from the support team. This was disappointing and frustrating for many customers. :-)

<p>For additional information, contact us at customerservice@technova-example.com !!!</p>

The price of the SmartView television is reasonable compared with similar products in the market. Many customers believe that the television provides good value for money. 💰💰

However, some customers feel that the product is slightly expensive for users with a limited budget. :-)

The product website is available at:
<https://www.technova-example.com/products/smartview>

Overall, the SmartView television is a good choice for customers who prioritize picture quality and streaming performance. 👍

The product has several impressive features, but the setup process and limited HDMI ports could be improved. Customers should consider their individual requirements before purchasing the television.

<p>Rating: ★★★★★/5</p>

<p>Thank you for sharing your feedback!!!</p>

Contact: feedback@technova-example.com

PRODUCT 2: VisionMax 55-inch Smart TV

<h1>Customer Feedback - VisionMax 55-inch Smart TV</h1>

<p>VisionMax has launched its latest 55-inch Smart TV, and customers have shared a variety of opinions. 😊</p>

The display quality is absolutely fantastic and the colors look bright and natural. 👍👍 The television has a very slim design and looks premium in the living room.

<p>Unfortunately, the television takes a long time to start. 😞 Sometimes users have to wait almost a minute before the home screen appears.</p>

Customers also complained that the operating system occasionally freezes while switching between applications. This is very frustrating!!! 😡

For support, contact support@visionmax-example.com or visit <https://www.visionmax-example.com/help>.

The built-in speakers produce clear sound, but the bass is not very powerful. Some customers recommended purchasing an external soundbar.

The television supports all major streaming platforms, and applications are easy to install. 😊 Netflix and YouTube work smoothly most of the time.

<p>However, some customers experienced application crashes while watching online content... 😞</p>

Gamers were generally satisfied with the low input lag. The television provides a smooth gaming experience and supports modern gaming consoles. 🎮

The number of HDMI ports is sufficient for most users, which is a major advantage.

<p>Customer service received mixed reviews.</p>

Some customers said that support representatives were polite and solved their problems quickly. 👍

Other customers complained that they had to explain the same issue multiple times before receiving a proper solution. :-(

The television is reasonably priced compared with similar products in the market.

However, customers with a limited budget may still find the price slightly expensive.

Overall, VisionMax provides excellent picture quality and a good gaming experience. However, the slow operating system and occasional application crashes are areas that need improvement.

Rating: ★★★★★/5

Contact: feedback@visionmax-example.com

PRODUCT 3: UltraView Pro 65-inch Smart TV

<h1>Customer Feedback - UltraView Pro 65-inch Smart TV</h1>

<p>The UltraView Pro 65-inch television has received extremely positive feedback from customers. 😍</p>

The large display is perfect for watching movies and sports. The picture quality is outstanding, and the colors are extremely accurate. 👍👍👍

Customers were also impressed with the brightness and contrast levels.

<p>The television has a premium metal design and feels very well built.</p>

The speakers provide excellent audio quality, although some customers felt that the maximum volume could be higher.

The television starts quickly and the operating system is very responsive. 😊

Streaming applications load almost instantly, and customers reported very few buffering issues.

<p>For technical assistance, users can contact support@ultraview-example.com.</p>

The television is also excellent for gaming. The low response time provides a smooth experience, and the television supports high refresh rates. 🎮

However, the television has only three HDMI ports. A few customers with multiple gaming consoles found this inconvenient. 😞

Customer service has received excellent reviews. Support representatives are friendly, knowledgeable, and quick to respond. 👍

Customers were especially happy with the quick replacement service when they received damaged products.

The biggest disadvantage is the price. The UltraView Pro is considerably more expensive than other televisions with similar screen sizes. 💰💰💰

Some customers believe the high price is justified because of the premium build and excellent performance.

Others feel that the product is too expensive for an average household.

<p>Overall, the UltraView Pro is an excellent television with outstanding

picture quality, fast performance, and excellent customer service.</p>

The high price is the only major concern reported by customers.

<https://www.ultraview-example.com/products/pro>

Rating: ★★★★★/5

Email: reviews@ultraview-example.com

PRODUCT 4: EcoScreen 50-inch Smart TV

<h1>Customer Feedback - EcoScreen 50-inch Smart TV</h1>

EcoScreen is a budget-friendly smart television designed for customers looking for an affordable entertainment option. 😊

The price is one of the biggest advantages of this television. Customers said that it offers excellent value for money. 👍

The television has a simple design and does not take up much space.

<p>The picture quality is acceptable for the price, but it is not as good as premium televisions.</p>

The colors sometimes look slightly dull, especially when watching movies in a dark room. 😞

Some customers also complained about poor viewing angles.

The built-in speakers are average. The sound is clear enough for regular television programs, but the bass is weak.

For questions, customers can contact help@ecoscreen-example.com.

The television supports popular streaming applications such as Netflix and YouTube. However, some applications take a long time to open.

<p>Customers reported occasional freezing while using the smart TV interface. 😞</p>

The television is not recommended for serious gamers because the response time is relatively high.

However, casual gamers were satisfied with the overall experience.

The customer support team received mixed feedback. Some customers received quick assistance, while others complained about slow responses.

One customer wrote that the support team took more than a week to resolve a simple technical issue. 🙄

The television is lightweight and easy to install.

Customers also appreciated the low electricity consumption, making it suitable for people who want to reduce their energy bills. 🌱

Overall, EcoScreen is a good option for customers who want an affordable smart television.

However, customers who prioritize picture quality, gaming performance, or premium sound may want to consider other products.

Rating: ★★☆☆/5

Visit: <https://www.ecoscreen-example.com>

Contact: support@ecoscreen-example.com

PRODUCT 5: CineMax Ultra 55-inch Smart TV

<h1>Customer Feedback - CineMax Ultra 55-inch Smart TV</h1>

The CineMax Ultra 55-inch Smart TV has received mostly positive reviews from customers. 😊

The picture quality is excellent, and the television produces deep blacks and vibrant colors. 👍👍

Customers who watch movies frequently were particularly impressed with the cinematic viewing experience.

<p>The television also supports Dolby Vision and HDR content.</p>

The design is modern and attractive, although some customers felt that the stand was not very stable.

One customer reported that the television accidentally tilted when the stand was

moved. 😞

The sound quality is very good for a built-in television speaker system.

However, customers who enjoy loud music may prefer an external sound system.

Streaming applications work smoothly, and the operating system is easy to navigate.

<p>Most applications open quickly, although a few users experienced occasional delays.</p>

For assistance, contact customerservice@cinemax-example.com.

The television performs reasonably well for gaming. 🎮 However, serious gamers complained about the lack of advanced gaming features.

The response time is acceptable, but it is not the best in this price range.

Customer service received excellent feedback. 👍

Customers appreciated the quick responses and helpful technical support.

One customer mentioned that a replacement remote control was delivered within two days.

The price is slightly higher than some competing televisions. 💰

However, most customers believe that the excellent picture quality justifies the price.

<p>Overall, CineMax Ultra is an excellent choice for movie lovers.</p>

It provides impressive picture quality, good sound, smooth streaming, and reliable customer service.

The main weaknesses are the average gaming performance and slightly expensive price.

Rating: ★★★★★/5

Website: <https://www.cinemax-example.com>

Email: support@cinemax-example.com

PRODUCT 6: SmartVision X 55-inch Smart TV

<h1>Customer Feedback - SmartVision X 55-inch Smart TV</h1>

<p>SmartVision X is a popular mid-range television that has received mixed customer reviews. 😞</p>

The television has a beautiful design and a very thin frame.

The picture quality is good, and the colors are bright. 👍

However, some customers reported that the screen brightness is not sufficient in rooms with strong sunlight.

<p>The television has excellent sound quality compared with other products in the same price range.</p>

The operating system is fast and easy to use.

Applications such as Netflix, YouTube, and Prime Video load quickly. 😊

Unfortunately, the television occasionally disconnects from Wi-Fi. 😞

Some customers said that they had to restart the television several times to reconnect to their home network.

For technical support, contact support@smartvision-example.com.

Gaming performance is average. Casual gamers were satisfied, but serious gamers complained about input lag. 😞

The television has four HDMI ports, which is sufficient for most households.

Customer service received positive feedback overall.

Support representatives were friendly and professional. 👍👍

However, a few customers complained that replacement parts were not always available immediately.

The price is competitive and affordable compared with similar televisions.

Many customers believe that the product provides good value for money.

<p>Some customers recommended the television for families because it is easy to use and offers good sound quality.</p>

However, users who require a highly reliable Wi-Fi connection may want to consider other options.

Overall, SmartVision X is a reasonably good television with good sound, attractive design, and competitive pricing.

The main issues are occasional Wi-Fi problems and average gaming performance.

Rating: ★★★★★/5

Visit: <https://www.smartvision-example.com>

Email: contact@smartvision-example.com

5. Task 1 – Read the Raw Input Data

Read the raw customer feedback data using Python.

You may:

- Store each product's feedback in a separate `.txt` file, OR
- Store all feedback in one text file, OR
- Store the feedback in a structured format such as CSV.

Recommended approach:

```
customer_feedback/  
|  
├─ technova.txt  
├─ visionmax.txt  
├─ ultraview.txt  
├─ ecoscreen.txt  
├─ cinemax.txt  
└─ smartvision.txt
```

The product name should be preserved because it will be required for the final product-wise analysis.

6. Task 2 – Create a Text Cleaning Function

Write a reusable Python function:

```
def clean_text(text):  
    """  
    Cleans raw customer feedback.  
  
    Parameters:  
        text (str): Raw input text  
  
    Returns:  
        str: Cleaned text  
    """  
  
    # Cleaning logic  
  
    return text
```

The function should programmatically clean the raw feedback.

6.1 Remove HTML Tags

Example:

```
<p>Hello customer</p>
```

Should become:

```
Hello customer
```

Remove tags such as:

```
<h1>  
<p>  
<b>  
<i>  
</p>  
</b>  
</i>
```

6.2 Remove Email Addresses

Example:

```
support@technova-example.com
```

Should be removed.

6.3 Remove URLs

Example:

```
https://www.technova-example.com/support
```

Should be removed.

6.4 Handle HTML Entities

Example:

```
bright & realistic
```

Should become:

```
bright & realistic
```

6.5 Handle Emojis

Remove emojis such as:





6.6 Remove Unwanted Special Characters

Identify and remove unnecessary special characters.

Examples:

```
:-(  
!!!  
...
```

Students should ensure that meaningful punctuation required for sentence boundaries is preserved where appropriate.

6.7 Normalize Multiple Spaces

Example:

The television has excellent picture quality.

Should become:

The television has excellent picture quality.

6.8 Normalize Newline Characters

Remove unnecessary blank lines and normalize multiple newline characters.

6.9 Normalize Repeated Punctuation

Example:

The product is excellent!!!

May be normalized to:

The product is excellent!

7. Task 3 – Save Cleaned Data

After cleaning each product's feedback, save the cleaned text.

Recommended structure:

```
cleaned_feedback/  
|  
├─ technova_cleaned.txt  
├─ visionmax_cleaned.txt  
├─ ultraview_cleaned.txt  
├─ ecoscreen_cleaned.txt  
├─ cinemax_cleaned.txt  
└─ smartvision_cleaned.txt
```

8. Task 4 – Perform Sentence Chunking

After cleaning the text, split the content into individual sentences.

Each sentence should be treated as one chunk.

Example:

Chunk 1:
TechNova Electronics has recently launched its new SmartView 55-inch television,
and customers have shared mixed opinions about the product.

Chunk 2:

The television has a modern design and looks very attractive in the living room.

Chunk 3:

The picture quality is excellent, and the colors are bright and realistic.

Students may use:

- NLTK
- Regular Expressions
- spaCy
- Other NLP libraries

The sentence chunking must be performed programmatically.

9. Task 5 – Assign Chunk IDs

Each sentence should have a unique ID.

Example:

Product	Chunk ID	Chunk Text
TechNova	1	TechNova Electronics has recently launched its new SmartView television...
TechNova	2	The television has a modern design...
VisionMax	1	VisionMax has launched its latest 55-inch Smart TV...

Students may either:

- Restart chunk IDs for each product, or
- Maintain a global unique chunk ID.

The `product` column must be maintained in either case.

10. Task 6 – Perform Sentiment Analysis

Perform sentiment analysis on every sentence chunk.

For every chunk, calculate:

- Sentiment score
- Sentiment label

The sentiment label should be:

- Positive
- Negative
- Neutral

Recommended approach:

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
```

Students should extract the VADER:

```
compound
```

score.

The compound score should be used to determine the sentiment label.

A commonly used threshold is:

```
compound >= 0.05      → Positive  
compound <= -0.05     → Negative  
-0.05 < compound < 0.05 → Neutral
```

11. Task 7 – Create Sentence-Level Output

Create a DataFrame containing all analyzed chunks.

The DataFrame should contain at least:

```
product  
chunk_id  
chunk_text  
sentiment_score  
sentiment_label
```

Example:

Product	Chunk ID	Chunk Text	Sentiment Score	Sentiment Label
TechNova	1	TechNova Electronics has recently launched...	0.20	Positive
TechNova	2	The television has a modern design...	0.70	Positive
TechNova	3	The installation instructions were not very clear...	-0.35	Negative
VisionMax	1	VisionMax has launched its latest television...	0.20	Positive

12. Task 8 – Save Sentence-Level Results

Save the complete analysis into:

```
customer_feedback_sentiment.csv
```

Required columns:

```
product
chunk_id
chunk_text
sentiment_score
sentiment_label
```

13. Task 9 – Generate Product-Wise Sentiment Summary

For each product, calculate:

1. Total number of chunks.
2. Number of positive chunks.
3. Number of negative chunks.
4. Number of neutral chunks.
5. Average sentiment score.
6. Positive percentage.
7. Negative percentage.

8. Neutral percentage.

The output should look similar to:

Product	Total Chunks	Positive	Negative	Neutral	Avg Score
TechNova	XX	XX	XX	XX	XX
VisionMax	XX	XX	XX	XX	XX
UltraView Pro	XX	XX	XX	XX	XX
EcoScreen	XX	XX	XX	XX	XX
CineMax Ultra	XX	XX	XX	XX	XX
SmartVision X	XX	XX	XX	XX	XX

14. Task 10 – Generate Sentiment Distribution

Calculate the percentage of each sentiment category for every product.

Example:

Product	Positive %	Negative %	Neutral %
TechNova	XX%	XX%	XX%
VisionMax	XX%	XX%	XX%
UltraView Pro	XX%	XX%	XX%
EcoScreen	XX%	XX%	XX%
CineMax Ultra	XX%	XX%	XX%
SmartVision X	XX%	XX%	XX%

15. Task 11 – Save Product-Wise Summary

Save the product-level analysis into:

product_sentiment_summary.csv

The file should contain:

```
product
total_chunks
positive_count
negative_count
neutral_count
positive_percentage
negative_percentage
neutral_percentage
average_sentiment_score
```

16. Task 12 – Identify the Best Product

Based on your analysis, identify the best Smart TV.

Consider:

- Highest average sentiment score
- Highest positive sentiment percentage
- Lowest negative sentiment percentage
- Product strengths
- Product weaknesses
- Recurring complaints
- Customer service feedback
- Price-related feedback
- Performance-related feedback

Do not select a product based on only one metric.

Provide a short business recommendation.

Example:

Recommended Product:

Average Sentiment Score:

Reason for Recommendation:

Key Strengths:

1.

- 2.
- 3.

Key Weaknesses:

- 1.
- 2.

Business Recommendation:

17. Task 13 – Business Insight

Write a short report answering:

Question 1

Which product received the most positive customer feedback?

Question 2

Which product received the most negative feedback?

Question 3

Which product has the highest average sentiment score?

Question 4

Which product has the lowest average sentiment score?

Question 5

Which product is best for:

- Movie lovers?
- Gamers?
- Budget-conscious customers?
- Families?
- Premium customers?

Question 6

Which product would you recommend overall?

Question 7

What are the major reasons behind your recommendation?

18. Task 14 – Optional Visualization

Create visualizations using Matplotlib or another visualization library.

Students may create:

Chart 1

Positive vs Negative vs Neutral sentiment by product.

Chart 2

Average sentiment score by product.

Chart 3

Number of positive and negative chunks per product.

Chart 4

Overall sentiment distribution.

19. Recommended Code Structure

Students are encouraged to create reusable functions.

At minimum:

```
def clean_text(text):  
    pass  
  
def sentence_chunk(text):  
    pass  
  
def analyze_sentiment(chunks):  
    pass
```



```
def create_product_summary(data):  
    pass  
  
def save_results(data, filename):  
    pass
```

A possible complete pipeline is:

```
# Step 1  
raw_text = read_input_file()  
  
# Step 2  
cleaned_text = clean_text(raw_text)  
  
# Step 3  
chunks = sentence_chunk(cleaned_text)  
  
# Step 4  
sentiment_results = analyze_sentiment(chunks)  
  
# Step 5  
save_results(  
    sentiment_results,  
    "customer_feedback_sentiment.csv"  
)  
  
# Step 6  
summary = create_product_summary(sentiment_results)  
  
# Step 7  
save_results(  
    summary,  
    "product_sentiment_summary.csv"  
)
```

20. Recommended Libraries

Students may use:

```
import re
import pandas as pd
import nltk
```

For sentiment analysis:

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
```

Optional libraries:

```
import spacy
import matplotlib.pyplot as plt
```

21. Expected Output Files

Students should submit the following:

```
NLP_Assessment/
|
|— code/
|   |— assessment.py
|
|— input/
|   |— technova.txt
|   |— visionmax.txt
|   |— ultraview.txt
|   |— ecoscreen.txt
|   |— cinemax.txt
|   |— smartvision.txt
|
|— cleaned/
|   |— technova_cleaned.txt
|   |— visionmax_cleaned.txt
|   |— ultraview_cleaned.txt
|   |— ecoscreen_cleaned.txt
|   |— cinemax_cleaned.txt
|   |— smartvision_cleaned.txt
|
|— output/
|   |— customer_feedback_sentiment.csv
```

```
|   └─ product_sentiment_summary.csv
|
└─ report/
    └─ business_recommendation.txt
```

22. Submission Requirements

Students should submit:

1. Python Source Code

```
assessment.py
```

or

```
assessment.ipynb
```

2. Raw Input Data

All six customer feedback files.

3. Cleaned Data

The cleaned version of all six feedback documents.

4. Sentence-Level Sentiment Analysis

```
customer_feedback_sentiment.csv
```

5. Product-Level Sentiment Summary

```
product_sentiment_summary.csv
```

6. Business Recommendation

A short report explaining:

- Which product is best.
- Why it is the best.

- Key strengths.
- Key weaknesses.
- Major customer complaints.
- Overall sentiment analysis.

7. Optional Visualizations

Submit any sentiment charts created.

23. Evaluation Criteria

Criteria	Marks
Reading raw input data	5
HTML tag removal	5
Email removal	5
URL removal	5
Emoji handling	5
Special character handling	5
Whitespace normalization	5
Sentence chunking	15
Chunk ID generation	5
Sentiment analysis	15
Sentence-level CSV generation	5
Product-wise aggregation	5
Business recommendation	5
Total	100

Bonus: 10 Marks

Students can earn up to 10 additional marks for:

- Creating sentiment visualizations.
- Identifying the most positive sentence.
- Identifying the most negative sentence.
- Extracting recurring customer complaints.
- Providing deeper business insights.
- Creating a product recommendation score.

24. Final Learning Outcome

After completing this assessment, students should be able to build a complete basic NLP pipeline:

```
Raw Text
↓
Data Cleaning
↓
Regex Processing
↓
HTML Removal
↓
Email Removal
↓
URL Removal
↓
Emoji Handling
↓
Special Character Handling
↓
Whitespace Normalization
↓
Sentence Tokenization
↓
Sentence Chunking
↓
Sentiment Analysis
↓
Sentiment Score
↓
Sentiment Label
↓
Pandas DataFrame
↓
CSV Export
↓
Product-Level Aggregation
↓
Sentiment Comparison
↓
Business Insights
↓
Product Recommendation
```

Final Business Question

You are the NLP/Data Analyst for a consumer electronics company. Based on the customer feedback from all six Smart TV products, which product would you recommend to customers and why? Support your answer with data from your sentiment analysis.

The goal of this assessment is not only to write Python code but also to demonstrate how **NLP techniques can be applied to solve a real-world business problem.**